

A Code-Based Post-Quantum Framework for Secure Aggregation in Federated Learning

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Abstract—Federated learning (FL) enables collaborative model training without sharing raw data, and *secure aggregation* protects the exchanged model updates by ensuring the server learns only their sum. The key establishment underlying these protocols is, however, vulnerable to quantum attacks, and the emerging body of post-quantum FL work derives its security almost exclusively from *lattice* assumptions. Concentrating an entire ecosystem on a single mathematical problem is a systemic risk: a future break of lattices would compromise all such systems at once. Motivated by NIST’s March 2025 selection of the code-based KEM *HQC* specifically to diversify away from lattices, we propose the first FL secure-aggregation framework whose post-quantum security comes from a code-based KEM rather than a lattice one. Because *HQC*’s ciphertexts are roughly four times larger and its operations orders of magnitude slower than lattice KEMs, we design a KEM-agnostic masking protocol with ciphertext-handling optimizations and honestly benchmark the resulting diversity-versus-cost trade-off—communication, computation, accuracy, and robustness—against a Kyber/ML-KEM baseline on public datasets.

Index Terms—Post-quantum cryptography, federated learning, secure aggregation, *HQC*, code-based cryptography, cryptographic diversity.

I. INTRODUCTION

Federated learning (FL) lets many clients train a shared model under a coordinating server while keeping their data local. Yet the model updates clients transmit can leak private training data, so *secure aggregation* ensures the server observes only the aggregate of all updates, never an individual one. Practical protocols establish pairwise secrets between clients, derive additive masks that cancel when updates are summed, and tolerate client dropout by secret-sharing the mask seeds. Their confidentiality rests on classical key exchange, which a cryptographically relevant quantum computer would break, retroactively exposing any recorded updates (“harvest-now, decrypt-later”).

The natural response is to replace the key exchange with a post-quantum KEM. Almost all post-quantum FL proposals, however, instantiate this with lattice-based cryptography [3]–[5], [7]. While lattices are efficient and now standardized as ML-KEM (FIPS 203) [2], a deployment-wide dependence on a single hardness assumption is fragile: a cryptanalytic advance against lattices would break every such system simultaneously. NIST acknowledged exactly this concern when,

in March 2025, it selected the code-based KEM *HQC*—whose security rests on syndrome decoding rather than lattices—to serve as a diversified backup to ML-KEM [1].

This paper asks whether that cryptographic diversity can be brought into federated learning. We propose, to the best of our knowledge, the first FL secure-aggregation framework built on a code-based KEM (*HQC*), and we characterize what the diversity costs. Our contributions are: (i) a *KEM-agnostic* secure-aggregation protocol whose key establishment can be instantiated with either *HQC* or a lattice KEM; (ii) ciphertext-handling techniques that keep per-round communication practical despite *HQC*’s much larger ciphertexts; and (iii) an honest, reproducible head-to-head evaluation of the code-based and lattice-based variants on public datasets. Preliminary measurements already quantify the gap: at comparable security, *HQC*’s ciphertext is 4433 B versus 1088 B for ML-KEM-768, and a key encapsulation/decapsulation costs ~ 34 ms versus ~ 0.17 ms—evidence that the swap is non-trivial rather than cosmetic.

II. RELATED WORK

Post-quantum FL relies almost exclusively on lattices. PQSF reconstructs masks with a lattice multi-stage secret-sharing scheme [3]; Qin and Xu achieve single-round aggregation using a lattice key-homomorphic pseudo-random function [4]; and Narsimhulu *et al.* aggregate CRYSTALS-Dilithium-style lattice signatures for low-latency distributed networks [5]. Non-lattice post-quantum FL is rare and has not used code-based KEMs: Singh *et al.*, for example, combine *multivariate*-cryptography-based security with blockchain for fog-edge FL [6]. This concentration mirrors standardization itself—the first standardized KEM and signature, ML-KEM (FIPS 203) [2] and ML-DSA, are both lattice-based—and recent surveys confirm lattices dominate post-quantum deployment [7]. Relying on one hardness assumption across an entire deployment is a systemic risk.

The missing axis: cryptographic diversity. In March 2025, NIST selected *HQC*, a *code-based* KEM, precisely to provide a backup whose hardness rests on syndrome decoding rather than lattices [1]. This diversity has not reached federated learning: to the best of our knowledge, no FL secure-aggregation framework uses a code-based KEM. We therefore present

what is, to our knowledge, the first FL secure-aggregation framework whose post-quantum security derives from the NIST-standardized code-based KEM (HQC), and the first to quantify the resulting diversity-versus-cost trade-off against a lattice (Kyber/ML-KEM) baseline.

III. PROPOSED FRAMEWORK

System and threat model. We consider standard cross-device/cross-silo FL with a server and N clients running FedAvg over non-IID data. The server is honest-but-curious and may collude with a bounded fraction of clients; we additionally consider malicious clients that submit poisoned updates and clients that drop out mid-round. The goal is that no coalition below a threshold learns any individual update, only the aggregate.

KEM-agnostic secure aggregation. Each client i generates a KEM keypair and publishes its public key. For each pair (i, j) , a shared secret is established by encapsulating to the peer's public key; a key-derivation function expands it into a pairwise pseudorandom mask m_{ij} that client i adds and client j subtracts. When the server sums all masked updates, the pairwise masks cancel and only the true aggregate remains. To survive dropouts, clients secret-share their seed material so the server can reconstruct and remove the masks of non-responding clients, following the established masking pattern. Crucially, the KEM is a pluggable module: the same protocol runs with the code-based HQC or the lattice-based Kyber/ML-KEM, isolating the cost of cryptographic diversity to a single component.

Making HQC practical. HQC's ciphertexts (e.g., 4433 B at NIST level 1) and slower operations would inflate per-round overhead if applied naively. We keep communication practical by amortizing key establishment across rounds (reusing long-lived KEM public keys and refreshing only masks), batching pairwise ciphertexts, and compressing transmitted key material, so the ciphertext overhead is paid once rather than every round.

Authentication scope. HQC is a KEM and cannot sign, and there is no standardized code-based signature. We therefore confine the code-based primitive to the confidentiality/key-establishment layer. Where update authentication is required, we use a separate standardized post-quantum signature (ML-DSA), stating explicitly that only the KEM layer is code-based; we do not attempt to "sign with HQC." The framework is implemented with liboqs (HQC and ML-KEM/Kyber), the Flower FL framework, and PyTorch, and will be released open-source.

IV. EVALUATION METRICS AND EXPERIMENTAL PLAN

We compare three configurations—plain FedAvg (no cryptography), lattice (Kyber/ML-KEM) secure aggregation, and code-based (HQC) secure aggregation—on three public datasets: FEMNIST and a non-IID partition of CIFAR-10 for image classification, and Edge-IIoTset for IoT intrusion detection, matching the security framing of related work. We report: (1) *communication*—bytes per round per client, separating

the model payload from key material and ciphertext overhead; (2) *computation*—keygen/encapsulation/decapsulation time and per-round wall-clock; (3) *utility*—test accuracy versus rounds, which must match the plain baseline since correct masking does not alter the aggregate; (4) *robustness*—accuracy under an increasing fraction of malicious/colluding clients and resilience to dropout; and (5) *security level*—the post-quantum strength (NIST levels 1/3/5) of each configuration. The headline result is the HQC-versus-Kyber comparison, presented as a transparent diversity-versus-cost trade-off; a quantified negative result (HQC too heavy for a given regime) is itself a useful finding.

V. CONCLUSION

Post-quantum federated learning today rests almost entirely on lattice cryptography, leaving it exposed to a single point of cryptanalytic failure. We propose the first FL secure-aggregation framework whose post-quantum security comes from a code-based KEM (HQC), turning NIST's diversification rationale into a concrete system, and we honestly measure what that diversity costs against a lattice baseline. By isolating the KEM behind a pluggable interface and releasing a reproducible implementation, the framework offers both a first code-based point of comparison for the community and a template for cryptographic agility in privacy-preserving machine learning.

REFERENCES

- [1] National Institute of Standards and Technology, "Status report on the fourth round of the NIST post-quantum cryptography standardization process," NIST, Gaithersburg, MD, USA, NIST IR 8545, Mar. 2025.
- [2] National Institute of Standards and Technology, "Module-lattice-based key-encapsulation mechanism standard," NIST, Gaithersburg, MD, USA, FIPS 203, Aug. 2024.
- [3] X. Zhang, H. Deng, R. Wu, J. Ren, and Y. Ren, "PQSF: Post-quantum secure privacy-preserving federated learning," *Sci. Rep.*, vol. 14, no. 1, Art. no. 23553, 2024, doi: 10.1038/s41598-024-74377-6.
- [4] X. Qin and R. Xu, "Efficient post-quantum cross-silo federated learning based on key homomorphic pseudo-random function," *Mathematics*, vol. 13, no. 9, Art. no. 1404, 2025, doi: 10.3390/math13091404.
- [5] P. Narsimhulu, P. Chithaluru, and R. Aluvalu, "Efficient post-quantum cryptographic signature aggregation for low-latency distributed networks," *EURASIP J. Inf. Secur.*, vol. 2026, Art. no. 6, 2026, doi: 10.1186/s13635-026-00228-8.
- [6] J. Singh, P. Singh, A. Kaur, and M. Hedabou, "Post-quantum secure fog-edge computing using federated learning with blockchain," *J. Supercomput.*, vol. 81, Art. no. 1249, 2025, doi: 10.1007/s11227-025-07738-1.
- [7] H. Nguyen, S. Huda, Y. Nogami, and T. T. Nguyen, "Security in post-quantum era: A comprehensive survey on lattice-based algorithms," *IEEE Access*, vol. 13, pp. 89003–89024, 2025, doi: 10.1109/ACCESS.2025.3571307.